characterized by

1. change in chemical and mineralogical composition
2. disappearance of some minerals
3. formation of new secondary minerals

Correct answer is

(i) 1 only
(ii) 1 and 2 only
(iii) 2 and 3 only
(iv) 1, 2 and 3

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(d) The difference between liquid limit and plastic limit is known as

(i) liquid limit and plastic limit
(ii) plastic limit and shrinkage limit
(iii) liquid limit and shrinkage limit
(iv) shrinkage limit and zero

(e) Wet sieve analysis is done for the soil passing

(i) 75 μm
(ii) 4·75 μm
(iii) 10 mm
(iv) 20 mm

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(h) Quick sand is

(i) a type of sand

(ii) a condition in which a cohesionless soil losses its strength because of upward flow of water

(iii) a condition in which a cohesive soil losses its strength

(iv) None of the above

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(iv) 47·8 kPa

2. (a) Explain the phenomena of formation and transportation of soils. 7

(b) Explain the method of determination of liquid limit of soil by Casagrande's apparatus. What are the limitations of the method? 7

3. (a) From fundamental, derive an expression for dry density in terms of bulk unit weight and water content. 7

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Flow index	8	10
Natural moisture content	40%	50%

(i) Which one is more plastic?

(ii) Which one is better foundation material when remoulded?

(iii) Which one has better strength as a function of water content?

(iv) Which one has better strength at the plastic limit? 7

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against piping? 6

(b) A silt deposit consists of two layers. The top layer is 2·5 m thick (Unit weight= 18 kN/m^3) and the bottom layer is 6·0 m thick (Unit weight = 19kN/m^3). The water table is at a depth of 4·0 m from the top. Draw the diagram showing the variation of total stress, neutral stress and effective stress. 8

7. (a) How the compaction improve the engineering properties of soil? 7

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of the soil over the base level of the foundation. 7

9. Write short notes on any *four* of the following : 14

(a) Black cotton soils

(b) Silt and clay

(c) Plasticity chart

(d) Piping

(e) Field compaction control

(f) Importance of pressure bulb

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